EVT in Large Core Infarction: PICO-Based Comprehensive Review

Expert Review | April 2026

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Sources: NEJM, Lancet Neurology, JAMA, Stroke

Abstract

Background: EVT in large core infarction was previously excluded. Six RCTs published 2022-2024 have established EVT as a reasonable treatment option.

Key Findings: 5 of 6 trials demonstrated significant benefit. **NNT = 6-13** for functional independence. No significant increase in symptomatic ICH.

📊 Understanding Outcome Measures

Outcome Type	Description	Trials
Binary mRS (0-3)	% achieving functional independence	RESCUE-JAPAN (primary)
Ordinal mRS Shift	Distribution shift across all 7 levels	ANGEL-ASPECT, SELECT2, TENSION
UW-mRS	Utility-weighted mean score (0-10, MCID=0.3)	TESLA
Median mRS	Median disability level at 90 days	LASTE

mRS Scale

Score	Description
0	No symptoms
1	No significant disability
2	Slight disability

Score	Description
3	Moderate disability
4	Moderately severe disability
5	Severe disability
6	Death

📋 PICO Summary Table

Trial	P	I	C	O
RESCUE-JAPAN LIMIT (2022)	n=203, LVO + ASPECTS 3-5, NIHSS ≥6, onset ≤6h	EVT + medical	Medical alone	**mRS 0-3: 36% vs 26%, P=0.02**
ANGEL-ASPECT (2023)	n=456, LVO + ASPECTS 3-5 or core 70-100 mL, onset ≤24h	EVT + medical	Medical alone	**mRS shift: OR 1.37, P=0.004**
SELECT2 (2023)	n=352, LVO + ASPECTS ≤5 or core ≥50 mL, onset ≤24h	EVT + care	Care alone	**mRS shift: OR 1.51, P<0.001**
TENSION (2023)	n=253, LVO + ASPECTS ≤5 or core ≥70 mL, onset 6-24h	EVT + therapy	Therapy alone	**mRS shift: OR 1.53, P=0.004**
TESLA (2024)	n=300, LVO + ASPECTS 2-5 (NCCT-only), onset ≤24h	EVT + usual	Usual alone	**UW-mRS: NS** ⚠️
LASTE (2024)	n=324, LVO + ASPECTS ≤5, NIHSS ≥6, onset ≤6.5h	EVT + therapy	Therapy alone	**mRS median: 4 vs 6, P<0.001**

🏥 Individual Trial Details

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1. RESCUE-JAPAN LIMIT (2022)

Study Design: Multicenter, Open-label, RCT, PROBE | **n=203**

Citation: Yoshimura S, et al. *N Engl J Med.* 2022;386:1303-1313. doi:10.1056/NEJMoa2118191

P (Population)

- **n = 203** (101 EVT, 102 Control)
- Acute ischemic stroke with **LVO (ICA or M1)**
- **ASPECTS 3-5** on NCCT or DWI-MRI
- **NIHSS ≥6**
- **Pre-stroke mRS 0-1**
- Age ≥18 years
- Onset ≤6h (or ≤24h if FLAIR negative)
- Randomization within 60 min of imaging

Exclusion: - Significant mass effect with midline shift - Acute intracranial hemorrhage - High hemorrhage risk (platelets <40,000, APTT >50 sec, PT-INR >3.0)

I (Intervention)

- **EVT:** Stent retriever, aspiration catheter, balloon angioplasty, or intracranial stent
- **IV alteplase:** 0.6 mg/kg (Japanese dose)

C (Control)

- Medical management alone (± IV alteplase if indicated)

O (Outcome)

Primary: mRS 0-3 at 90 days

Outcome	EVT (n=101)	Control (n=102)	OR (95% CI)	P-value	NNT
mRS 0-3 (Primary)	**36%**	**26%**	**—**	**P=0.02**	**10**
mRS 0-2 at 90d	9.5%	0%	—	—	—
mRS shift (Secondary)	—	—	**OR 1.42 (0.88-2.29)**	**P=0.15 (NS)**	—
Symptomatic ICH	9%	5%	—	P=0.24	—

> ✅ Primary outcome significant, secondary (mRS shift) not significant

2. ANGEL-ASPECT (2023)

Study Design: Multicenter, Prospective, Open-label, RCT | **n=456** (230 EVT, 226 Control)

Citation: Li Q, et al. *N Engl J Med.* 2023;388:1240-1251. doi:10.1056/NEJMoa2213379

P (Population)

- **n = 456** (46 centers, China)
- **LVO (ICA or M1)**
- **ASPECTS 3-5** or **core 70-100 mL** on CTP or DWI-MRI
- Age ≥18 years
- **Onset ≤24 hours**

Exclusion: - Pre-stroke mRS >2 - Intracranial hemorrhage - Mass effect with midline shift >5 mm - Platelet <100,000 or anticoagulant use - Creatinine >1.5x upper limit

I (Intervention)

- **EVT + Medical management** (standard care)
- IV alteplase if indicated

C (Control)

- Medical management alone

O (Outcome)

Primary: Ordinal mRS shift at 90 days

Outcome	EVT (n=230)	Control (n=226)	OR (95% CI)	P-value	NNT
mRS shift (Primary)	—	—	**1.37 (1.11-1.69)**	**P=0.004**	**8**
mRS 0-3 at 90d	20%	12%	—	P=0.02	12.5
Reperfusion at 24h (TICI ≥2b)	52%	27%	—	—	—
Symptomatic ICH	6.1%	2.7%	—	P=0.07	—
90-day mortality	22%	21%	—	NS	—

3. SELECT2 (2023)

Study Design: International, Multicenter, Prospective, Open-label, Adaptive RCT | **n=352** (stopped early)

Citation: Sarraj A, et al. *N Engl J Med.* 2023;388:1252-1262. doi:10.1056/NEJMoa2214403

P (Population)

- **n = 352** (178 EVT, 174 Control)
- **LVO (ICA or M1)**
- **ASPECTS 3-5** or **core ≥50 mL** on CTP or DWI-MRI
- Age 18-85 years
- **Onset ≤24 hours**

Exclusion: - Pre-stroke mRS >2 - Intracranial hemorrhage - Age >85 with ASPECTS <5

I (Intervention)

- **EVT + Medical care** (stent retrievers, aspiration catheters)

C (Control)

- Medical care alone

O (Outcome)

Primary: mRS at 90 days (ordinal shift)

Outcome	EVT (n=178)	Control (n=174)	OR (95% CI)	P-value	NNT
mRS shift (Primary)	—	—	**1.51 (1.20-1.89)**	**P<0.001**	**4.7**
mRS 0-3 at 90d	**30%**	**15%**	**RR 2.06**	—	**6.7**
mRS 0-2 at 90d	18%	7%	RR 2.97	—	9
Symptomatic ICH	1%	2%	—	—	—

4. TENSION (2023)

Study Design: Multicenter, Open-label, RCT | **n=253**

Citation: Bendszus M, et al. *Lancet Neurol.* 2023.

P (Population)

- **n = 253**
- **LVO (ICA or M1)**
- **ASPECTS ≤5** or **core ≥70 mL**
- Age ≥18 years
- **Onset 6-24 hours**

Exclusion: - Pre-stroke mRS >2 - Intracranial hemorrhage - Platelets <50,000

I (Intervention)

- **EVT + Medical therapy**

C (Control)

- Medical therapy alone

O (Outcome)

Primary: Ordinal mRS shift at 90 days

Outcome	EVT	Control	OR (95% CI)	P-value
mRS shift (Primary)	—	—	**1.53 (1.14-2.04)**	**P=0.004**
mRS 0-2 at 90d	29%	17%	—	—
Symptomatic ICH	Similar	Similar	—	NS

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5. TESLA (2024) ⚠️ NEGATIVE TRIAL

Study Design: Multicenter, Open-label, Blinded-endpoint, Bayesian-adaptive RCT | **n=300** (152 EVT, 148 Control)

Citation: Costalat V, et al. *JAMA.* 2024.

P (Population)

- **n = 300** (152 EVT, 148 Control)
- **LVO (ICA or M1)**
- **ASPECTS 2-5 on NONCONTRAST CT ONLY** (no perfusion imaging)
- Age ≥18 years
- **Onset ≤24 hours**

Exclusion: - Core >100 mL - Pre-stroke mRS >2

I (Intervention)

- **EVT + Usual care** (standard thrombectomy devices)

C (Control)

- Usual care alone

O (Outcome)

Primary: Utility-weighted mRS at 90 days

Outcome	EVT (n=152)	Control (n=148)	Effect	P-value	Result
UW-mRS (Primary)	**2.93**	**2.27**	**Diff 0.63 (95% CrI -0.09 to 1.34)**	**Post. prob 0.96**	⚠️ **NOT SIGNIFICANT**
90-day mortality	35.3%	33.3%	—	NS	—
Symptomatic ICH	4.0%	1.3%	—	—	—

> ⚠️ 95% CrI included both no effect AND clinically relevant benefit. NCCT-only selection may be insufficient.

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6. LASTE (2024)

Study Design: Multicenter, Randomized, Open-label, Adjudicator-blinded | **n=324** (159 EVT, 165 Control)

Citation: Moret C, et al. *N Engl J Med.* 2024;390:1677-1689. doi:10.1056/NEJMoa2314063

P (Population)

- **n = 324** (159 EVT, 165 Control)
- **LVO (ICA or M1)**
- **ASPECTS ≤5** on NCCT or DWI (>3 if age >80)
- **NIHSS ≥6** (median: 21)
- Pre-stroke mRS ≤1
- **Onset ≤6.5 hours** (or ≤24h if FLAIR negative)
- Median ASPECTS: 2

Exclusion: - Intracerebral hemorrhage or midline shift - Cervical ICA disease requiring stent - Creatinine >4.0 mg/dL

I (Intervention)

- **EVT + Medical therapy**

- Median time from onset to thrombectomy: 305 minutes

C (Control)

- Medical therapy alone

O (Outcome)

Primary: Median mRS score at 90 days

Outcome	EVT (n=159)	Control (n=165)	OR (95% CI)	P-value	NNT
mRS median (Primary)	**4**	**6**	**1.63 (1.29-2.06)**	**P<0.001**	**~7**
All-cause death 90d	**36.1%**	**55.5%**	**RR 0.65 (0.50-0.84)**	**P<0.001**	**5**
Symptomatic ICH	9.6%	5.7%	RR 1.73 (0.78-4.68)	NS	—

> ✅ First trial to show significant mortality reduction (NNT=5 for survival)

📊 Meta-Analysis (6 Trials, n=1,888)

Outcome	NNT	95% CI
Better mRS shift	4.7	3.7-6.6
Functional independence (mRS 0-2)	7.1	5.6-9.6
Independent walking (mRS 0-3)	10.6	8.2-14.8

Safety	Risk Diff	95% CI	P-value
Symptomatic ICH	+1.7%	-0.5 to +3.9%	**0.10 (NS)**

📋 2026 AHA/ASA Guideline

Recommendation	Class	Level
EVT for LVO + large core (ASPECTS 3-5), ≤24h	**Class 2a**	**A**
May use NCCT alone	Class 2b	B

Topic	2019	2026
Large Core EVT	Not recommended	**Class 2a, Level A**
Imaging	Required CTP/MRI	May use NCCT alone

Topic	2019	2026
Time Window	6 hours	Up to 24 hours

💡 Take-home Points

1. **5 of 6 trials showed significant benefit**
2. **TESLA was the only negative trial** — NCCT-only may be insufficient
3. **NNT = 6-13** for functional independence
4. **No significant increase in symptomatic ICH** (P=0.10)
5. **LASTE showed mortality reduction** — RR 0.65, NNT=5

📚 References

1. Yoshimura S, et al. RESCUE-JAPAN LIMIT. *N Engl J Med.* 2022;386:1303-1313. doi:10.1056/NEJMoa2118191

2. Li Q, et al. ANGEL-ASPECT. *N Engl J Med.* 2023;388:1240-1251. doi:10.1056/NEJMoa2213379

3. Sarraj A, et al. SELECT2. *N Engl J Med.* 2023;388:1252-1262. doi:10.1056/NEJMoa2214403

4. Bendszus M, et al. TENSION. *Lancet Neurol.* 2023.

5. Costalat V, et al. TESLA. *JAMA.* 2024.

6. Moret C, et al. LASTE. *N Engl J Med.* 2024;390:1677-1689. doi:10.1056/NEJMoa2314063

7. Chen H, et al. Meta-analysis of 6 RCTs. *J Neurointerv Surg.* 2025. doi:10.1136/jnis-2023-021366

8. 2026 AHA/ASA Guideline. *Stroke.* 2026.

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All data from original papers following PICO format